KAIZEN MATHS

Lesson starter

Friction and Rough Surfaces

Friction Basics

Name _____

S1

A block of mass 8 kg rests on a horizontal rough surface. Find the normal reaction. Take g = 9.8 m s^{-2}.

S2

A block of mass 4 kg rests on a rough horizontal surface. The coefficient of friction is 0.3. Find the limiting friction. Take g = 9.8 m s^{-2}.

S3

A block of mass 2 kg is on a horizontal rough surface. The limiting friction is 3.92 N. Find the coefficient of friction. Take g = 9.8 m s^{-2}.

S4

A horizontal force of 9 N is applied to a block of mass 4 kg on a rough horizontal surface. The coefficient of friction is 0.4. Decide whether the block slides. Take g = 9.8 m s^{-2}.

Show your working

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MEC-DED4-L1 · S

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KAIZEN MATHS

Exit ticket

Friction and Rough Surfaces

Friction Basics

Name _____

E1

The coefficient of friction between two surfaces is 0.3. Find the angle of friction to the nearest degree.

E2

A block of mass 20 kg rests on a horizontal rough surface. Find the normal reaction. Take g = 9.8 m s^{-2}.

E3

A block of mass 15 kg rests on a rough horizontal surface. The coefficient of friction is 0.3. Find the limiting friction. Take g = 9.8 m s^{-2}.

KAIZEN MATHS

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Friction and Rough Surfaces

Friction Basics

Name _____

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The coefficient of friction between two surfaces is 0.3. Find the angle of friction to the nearest degree.

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A block of mass 20 kg rests on a horizontal rough surface. Find the normal reaction. Take g = 9.8 m s^{-2}.

E3

A block of mass 15 kg rests on a rough horizontal surface. The coefficient of friction is 0.3. Find the limiting friction. Take g = 9.8 m s^{-2}.

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Friction and Rough Surfaces

Friction Basics

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The coefficient of friction between two surfaces is 0.3. Find the angle of friction to the nearest degree.

E2

A block of mass 20 kg rests on a horizontal rough surface. Find the normal reaction. Take g = 9.8 m s^{-2}.

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A block of mass 15 kg rests on a rough horizontal surface. The coefficient of friction is 0.3. Find the limiting friction. Take g = 9.8 m s^{-2}.

MEC-DED4-L1 · E

Show your working

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Exit ticket

Friction and Rough Surfaces

Friction Basics

Name _____

E1

The coefficient of friction between two surfaces is 0.3. Find the angle of friction to the nearest degree.

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A block of mass 20 kg rests on a horizontal rough surface. Find the normal reaction. Take g = 9.8 m s^{-2}.

E3

A block of mass 15 kg rests on a rough horizontal surface. The coefficient of friction is 0.3. Find the limiting friction. Take g = 9.8 m s^{-2}.

MEC-DED4-L1 · E

Show your working

Teacher answers & working

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S1

A block of mass 8 kg rests on a horizontal rough surface. Find the normal reaction. Take g = 9.8 m s^{-2}.

Answer:

78.4 N

Working:

On a horizontal surface with no other vertical forces, the normal reaction balances the weight.

$R = mg$

$R = 8 \times 9.8$

$R = 78.4\ N$

S2

A block of mass 4 kg rests on a rough horizontal surface. The coefficient of friction is 0.3. Find the limiting friction. Take g = 9.8 m s^{-2}.

Answer:

11.76 N

Working:

Find the normal reaction.

$R = mg = 4 \times 9.8 = 39.2\ N$

Limiting friction is the maximum possible friction before sliding begins.

$F = \mu R$

$F = 0.3 \times 39.2$

$F = 11.76\ N$

S3

A block of mass 2 kg is on a horizontal rough surface. The limiting friction is 3.92 N. Find the coefficient of friction. Take g = 9.8 m s^{-2}.

Answer:

$\mu = 0.2$

Working:

Find the normal reaction.

$R = mg = 2 \times 9.8 = 19.6\ N$

Use limiting friction.

$F = \mu R$

$3.92 = \mu \times 19.6$

$\mu = \frac{3.92}{19.6}$

$\mu = 0.2$

Teacher answers & working

Friction and Rough Surfaces · Friction Basics

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S4

A horizontal force of 9 N is applied to a block of mass 4 kg on a rough horizontal surface. The coefficient of friction is 0.4. Decide whether the block slides. Take g = 9.8 m s^{-2}.

Answer:

The block does not slide.

Working:

Find the maximum possible static friction.

$R = mg = 4 \times 9.8 = 39.2$ N

$F_{max} = \mu R = 0.4 \times 39.2 = 15.68$ N

Compare the applied force with the maximum friction.

$9 \leq 15.68$

The applied force can be balanced by static friction, so the block does not slide.

E1

The coefficient of friction between two surfaces is 0.3. Find the angle of friction to the nearest degree.

Answer:

$17°$

Working:

The angle of friction λ satisfies $\tan \lambda = \mu$.

$\tan \lambda = 0.3$

$\lambda = \tan^{-1}(0.3)$

$\lambda = 17°$

E2

A block of mass 20 kg rests on a horizontal rough surface. Find the normal reaction. Take g = 9.8 m s^{-2}.

Answer:

196 N

Working:

On a horizontal surface with no other vertical forces, the normal reaction balances the weight.

$R = mg$

$R = 20 \times 9.8$

$R = 196$ N

Teacher answers & working

Friction and Rough Surfaces · Friction Basics

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E3

A block of mass 15 kg rests on a rough horizontal surface. The coefficient of friction is 0.3. Find the limiting friction. Take $g = 9.8$ m s^{-2}.

Answer:

44.1 N

Working:

Find the normal reaction.

$R = mg = 15 \times 9.8 = 147$ N

Limiting friction is the maximum possible friction before sliding begins.

$F = \mu R$

$F = 0.3 \times 147$

$F = 44.1$ N